Commun Nonlinear Sci Numer Simulat 130 (2024) 107768

Contents lists available at ScienceDirect

Communications in Nonlinear Science and Numerical Simulation

journal homepage: www.elsevier.com/locate/cnsns

Research paper

Revisiting 'survival of the fittest' principle in global stochastic optimisation: Incorporating anisotropic mutations

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ARTICLE INFO

Keywords:
Global stochastic optimisation
Convergence rate
Evolutionary algorithms
Natural selection, optimal control of infections

ABSTRACT

Evolutionary methods of global stochastic optimisation are very popular in the literature with various practical applications, including modelling of natural disasters. To enhance efficiency of optimisation, advanced algorithms seek to incorporate the heterogeneity of population (trial points) and consider more complex scenarios of mutations and recombination. On the other hand, most of the existing evolutionary algorithms have the major drawback that they cannot guarantee the convergence to the global extremum, in other words, such algorithms are purely heuristic. Here we proposed a new approach to global stochastic optimisation using evolutionary algorithms, where we can rigorously show convergence to the optimal solution with probability of one. This was a challenge for previous heuristic algorithms of optimisation. We provide the proof of convergence of the novel algorithm both for the case, where the parameter space has a finite dimensionality, and for optimisation in function spaces, with infinite number of dimensions. The crux of the proposed algorithms is to describe optimisation as a long-term natural selection process, which is a result of interplay between competition for limited resources and mutation, which is known as survival of the fittest principle. This is a modification of Monte-Carlo optimisation, where at each iteration, the search is refined by generating mutations to eventually attain the neighbourhood with the maximal population fitness. Unlike the previous algorithms, using survival of the fittest philosophy, we incorporate the anisotropic mutation for different spatial coordinates. We also suggest implementation of combination of the local and the global search. Using various multi-modal test functions in high dimensional spaces, we show that the new optimisation methods can demonstrate a better performance than some other well-known bio-inspired optimisation algorithms. As an important practical application, we implement our method to find the optimal control strategy in a multi-agent model of spread of infectious diseases. We argue that incorporating anisotropy in mutations is a promising direction in constructing more efficient evolutionary methods of optimisation, which has not been properly exploited in previous algorithms yet.

1. Introduction

Global optimisation problems arise in many areas of research, such as economics, finance, physics and modelling natural disasters. Generally, there is a need for quick, reliable and low cost algorithms for solving optimisation problems in parameter spaces of high (and even infinite) dimensionality with applications to systems with artificial intelligence, neural networks technologies, as well as

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https://doi.org/10.1016/j.cnsns.2023.107768

Available online 7 December 2023
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machine learning [1–4]. A large number of methods of global stochastic optimisation apply the so-called nature-inspired algorithms, in particular, bio-inspired algorithms, which mimic various biological processes [5,6]. Bio-inspired optimisation largely implements ideas from evolution, adaptation, population genetics, and animal behaviour [7]. Optimisation methods imitate long-term biological evolution and natural selection, and they largely utilise the phenomena of mutations and recombination (crossover) [8–11].

Currently, evolutionary algorithms of global optimisation are actively being developed, and among major directions are the following ones: (i) incorporating the heterogeneity of the population of searching agents, and (ii) better mimicking the processes of mutation and crossover, observed in natural systems. The idea of heterogeneity of the population of searchers is borrowed from algorithms of swarm optimisation [7,11,12]. The whole population of searchers is split into two or more subpopulations: one subpopulation (explorers or scouts) performs search for the optimal value across the global parameter space, whereas other subpopulations (exploiters) improve results achieved within previous iterations at some promising local subdomains [13].

To better mimic mutations in evolutionary algorithms, one can account for the heterogeneity of space of search, by using distinct mutation rates for different spatial domains [14]. Other promising approaches consider condition-dependent mutations. Mathematically, this signifies that different mutation operators are applied in an adaptive way to accelerate the search for the optimal solution [9,15]. The choice of the form of the mutation operator can be made using information on solving similar optimisation problems previously, from an external archive [16]. Modern approaches reconsider the phenomenon of crossover in genetics and this results in more complicated mutation operators [15,17,18]. The parameters, describing the strength of mutation, can be chosen to follow a harmonic function [19]: one can implement a harmonic function with either increasing or decreasing amplitude, as the number of generations increases (the frequency of the function is assumed to be a random variable). Another productive idea is constructing the so-called opposite points. In this case, after generating mutation points, the set of opposite points is constructed, so those points are symmetric with respect to the middle point of the interval; the new population is formed using the combined set of the initially generated and the opposite points [20].

However, the vast majority of existing evolutionary optimisation algorithms are still *heuristic*. In other words, a rigorous demonstration of convergence of these methods to the global solution is still to be proved. This is clearly a major issue for implementation of such methods. Indeed, even if a particular algorithm has demonstrated a number of previously successful implementations, there is no guarantee that the considered method would find the optimal solution for a new class of problems. Unfortunately, different methods of global optimisation, applied to the same problem, may provide different solutions (see examples in [21]). This is because heuristic methods can get stuck in the vicinity of some local (i.e. non-global) maxima. Clearly, this is a serious obstacle for practical applications of heuristic methods. Finally, we should stress that in most algorithms of evolutionary optimisation, the process of mutations is still incorporated in a rather simple way, in particular assuming mutation rates to be isotropic across the search space.

A promising solution to address the above problems in stochastic optimisation might be implementation of the famous survival of the fittest principle in biology postulated by Charles Darwin [22–24]. The main idea is to imitate biological evolution in optimisation by considering mutation and natural selection within a population of interacting species competing for limited resources: in ecology, this scenario is related to the Verhulst modelling approach resulting in the emergence of a carrying capacity of the environment [25]. The fittest genotype, i.e. the one to eventually survive and outcompete the others, will correspond to the maximum of the objective function. Using this straightforward paradigm, a novel evolutionary optimisation algorithm has been created, Survival of the Fittest Algorithm (SoFA) [24]. The SoFA mimics the processes of natural selection, due to the reproduction success of the population, and production of mutants described by the Gaussian or Cauchy probability density function [24,26]. The algorithm has been rigorously demonstrated to possess convergence in probability of one to the point of the global maximum and has been successfully implemented for solving the problems of optimal control of heat transfer [27], metal rod deformations [28], training of neural networks [29], or revealing evolutionarily optimal strategies in structured populations [24]. On the other hand, the rate of convergence for SoFA can be slower than for some known heuristic optimisation algorithms [26], which may limit the practicality of this method.

In this study, we show that the optimisation using survival of the fittest paradigm has a great potential and can be largely enhanced in terms of achieving high convergence rates. Importantly, this can be done via introducing an anisotropic operator of mutations, which can be considered as an analogue of recombination (crossover) in genetics. We stress that allowing for anisotropic mutations seems to be a completely new and promising idea in the entire research field of evolutionary optimisation. In the algorithms proposed here, we also implement the ideas of combining global and local optimisation, which has a similar philosophy as the swarm optimisation using heterogeneous population of searchers. In this study, we suggest several new global optimisation algorithms by incorporating the above mentioned ideas. For the newly created algorithms, we rigorously prove their convergence. We show that the novel algorithms demonstrate a greater performance as the original SoFA. Moreover, comparison with other well-known stochastic global optimisation techniques including ESCH [30], CRS [31,32], and differential evolution, DE (the 'current-to-best' version) [33] demonstrate a high competitiveness of our approach in terms of their high convergence rate on various classes of multi-modal functions. We also demonstrate that the new method can be efficiently applied to reveal the optimal control strategy of disease spread in a multi-agent model.

The paper is structured as follows. In Section 2, we introduce the basic algorithm as well as its modifications and construct a family of new methods. In Section 3, we rigorously prove the convergence of the basic algorithm. In Section 4 we compare the efficiency of the proposed methods with some previous global optimisation methods; we also explore the effect of particular parameterisation of operator on the optimisation speed. Section 5 provides an example of how our optimisation methodology can be applied to find the optimal control solution in a multi-agent model of infection disease spread. The Discussion section (Section 6) provides further ideas of improving the method and emphasize the links between the novel optimisation algorithms and biological evolution and selection. The Summary concludes the paper.

2. Methodology

2.1. Problem statement

We consider the optimisation problem in the m-dimensional box $\Pi = \{z = (x_1, x_2, \dots, x_m) : a_i < x_i < b_i, i = 1, \dots, m\}$, where a certain positive continuous function $J(z)$ is defined (the positivity of $J(z)$ can be always assumed: if it is not the case, one can use a simple transform to ensure $J(z) > 0$ in Π). We assume that $J(z)$ has a unique global maximum at the point z^*, which is unknown. The optimisation problem consists in finding the point z^* with some specified accuracy.

To solve the above mentioned problem, we propose the optimisation algorithm, whose basic version is stated below (Section 2.2). To improve the rate of convergence for the basic algorithm, several extensions (modifications) are listed in Section 2.3.

2.2. Description of the basic algorithm

The proposed optimisation algorithm uses two numerical sequences: ϕ_k and σ_k, such that $\lim_{k\to\infty} \phi_k = \infty$ and $\lim_{k\to\infty} \sigma_k = 0$. The choice of the above sequences regulates the convergence rate of the optimisation. The other parameters of the method are: a real number $Q \in [0, 1]$, and a natural number $M > 1$.

Firstly, we need to initialise the method (considering this as a 'zero' iteration, $k = 0$), by generating a uniformly distributed set of points $z_{01}, z_{02}, \dots, z_{0n}$ in the box Π, and we evaluate the function $J(z)$ at each point: $J(z_{01}), J(z_{02}), \dots, J(z_{0n})$. Note that for all other iterations, the number of the points n, where the function J is being evaluated, remains constant. In this case, we can talk about a set of interacting genotypes (individuals), constituting the global population, with the total size n remaining constant (n can be understood as the carrying capacity of the population of some interacting individuals). Therefore, n can be considered as a parameter for the proposed optimisation procedure.

Next, at each iteration $k + 1$ ($k \geq 0$), we renew the population of points based on the previous configuration from the iteration k, i.e. using the points $z_{k,1}, z_{k,2}, \dots, z_{k,n}$ with the corresponding values $J(z_{k,1}), J(z_{k,2}), \dots, J(z_{k,n})$. To make this renewal, we perform the following operations.

1. Choose a reference point. We select a reference point $z' = (x'_1, x'_2, \dots, x'_m)$ out of the present points $z_{k,1}, z_{k,2}, \dots, z_{k,n}$ with the probability defined by the ratio $J^{\phi_k}(z_{k,i})/(J^{\phi_k}(z_{k,1}) + J^{\phi_k}(z_{k,2}) + \dots + J^{\phi_k}(z_{k,n}))$, where $i = 1, \dots, n$. Biologically, this procedure reflects competition of genotypes (individuals) within the population of the size n, where the success of competition is related to the reproduction rate (the number of offspring), which, in turn, is expressed by the fitness function $J(z)$.
2. Use the obtained reference point z', we choose the point z_{k+1} for a further test. The jth component of this point is randomly chosen from the interval $[a_j, b_j]$ with the probability defined by the normal law with the probability density function $A_{k+1,j} \exp(-(x_j - x'_j)^2/(s_{k+1,j}))$. Here, $s_{k+1,j}$ is the variance of the considered normal distribution divided by 2. We set $s_{k+1,j}$ as follows. We choose a uniformly distributed random number q_j from the interval $[0, 1]$, and we compare q_j with the parameter Q: in the case $q_j < Q$, then $s_{k+1,j} = \sigma_{k+1}$ (strong mutation), otherwise, we set $s_{k+1,j} = \sigma_{k+1+M}$ (weak mutation). The sequence σ_k is defined above. The constants $A_{k+1,j}$ are chosen to normalise the probabilities, so integration of the probability density function over $[a_j, b_j]$ should give 1. Therefore, the point z_{k+1} with m components is randomly chosen from Π, with the probability described by the m-dimensional normal distribution given by:
$$A_{k+1} \exp\Big(-\frac{(x_1 - x'_1)^2}{s_{k+1,1}} - \frac{(x_2 - x'_2)^2}{s_{k+1,2}} - \dots - \frac{(x_m - x'_m)^2}{s_{k+1,m}}\Big), \tag{1}$$
where the constant A_{k+1} is found from normalisation by integrating the probability density function over Π. One can re-write the above expression in a matrix form as:
$$A_{k+1} \exp\Big(-(z_{k+1} - z'_{k+1})\Sigma^{-1}_{k+1}(z_{k+1} - z'_{k+1})^T\Big), \tag{2}$$
where Σ_{k+1} is the diagonal matrix with the diagonal elements of $s_{k+1,j}$, and Σ^{-1}_{k+1} is the corresponding inverse matrix.
From the biological point of view, this process describes generation of mutations within the population, or, more precisely, those mutations produced by the genotype z'. Note that unlike the existing evolutionary optimisation methods, here we use an anisotropic operator to model mutations: the rate of mutation is distinct for distinct hereditary traits. This scenario of mutations is somewhat similar (although not being fully equivalent) to the phenomenon of the crossover in genetics, where the offspring modify only some of hereditary traits.
3. The function J is evaluated at the newly found point z_{k+1}. In the case, where $J(z_{k+1}) > J(z')$, the reference point is replaced by z_{k+1}, and the population of n points becomes updated. Biologically, this corresponds to the process of natural selection within the population, where the genotype with a higher fitness J would be selected and will remain within the population.

The method terminates based on either of the following criteria: (a) the maximal possible number of iterations is achieved, or (b) the probability density distribution to generate the search point at the next iteration approaches the δ- function with the centre corresponding to the maximal value of J. In this study, for the sake of simplicity and also for comparison with the existing optimisation algorithms, we mostly use the fixed number of iterations as the termination criterion. However, we argue that implementing criterion (b) can provide a novel direction in construction of optimisation algorithms, which is discussed in Section 6.

The proof of the convergence of the above algorithm is provided in Section 3.

2.3. Modifications of the basic algorithm

The above basic optimisation algorithm allows various modifications, which can substantially improve the convergence rate of the method. Each modification can be formally considered as a distinct method, although related to the basic algorithm. Note that for each proposed modification, the property of convergence is conserved, and this can be proved analytically (the corresponding proofs are similar to the one provided for the basic algorithm, therefore, they are not provided here for brevity). As such, using the basic algorithm, one can construct a family of efficient optimisation algorithms, which demonstrates its flexibility. The considered here modifications involve the following three central ideas listed below and their combinations.

Idea 1. We combine the global and the local optimisation approaches. Such combination can be realised as follows. The function J is evaluated at the point z_k, as described in the basic algorithm. However, if the value $J(z_k)$ is high enough, we implement a local rather than a global search of the next iteration point. In particular, we consider a point z_k to be suitable for local optimisation search in the case: $J(z_k) > \xi J_{max}$, where J_{max} is the maximal value of J after iteration k, ξ is a tuning parameter $\xi \in [0,1]$. The local search for the next interaction point is done by estimating the gradient of J at the point z_k. We estimate the components of the gradient of J as $G_j = (J(z_k + dyI_j) - J(z_k))/dy$, where dy is a small increment, I_j is a unit vector directed along the j th spatial component, $j = 1, 2, \dots, m$. The search of the next point of iteration is performed along the vector $(G_1, G_2, \dots, G_m)$. We introduce a small distance d, and consider three points for further tests, having distances from z_k given by d, $d/2$, and $2d$. At each of those three points, we evaluate the function J, and then select the one with the highest J. This will be the point z_{k+1}.

Idea 2. We apply Idea 1, but to define the length of the local shift d, we use the local estimate of the Lipschitz constant L in the vicinity of the point z_k obtained at iteration k. We estimate the Lipschitz constant as $L = \max_j |G_j|$, where G_j was defined in the description of Idea 1. The estimate L is then used to find the length of the shift d by setting $d = sL$, where s is some fixed constant.

Idea 3. We replace the probability density function, $A_{k,j} \exp(-(x_j - x'_j)^2/s_{k,j})$, introduced in the mutation operator of the basic algorithm, with the following one given by

$$A'_{k,j}(x_j - x'_j)^2 \exp(-(x_j - x'_j)^2/s_{k,j}), \tag{3}$$

where $A'_{k,j}$ is a normalising constant. It is easy to see that function (3) vanishes at z'. Therefore, mutations are rare within a close vicinity of the reference point, and the maximal probability density of mutations occurs at some distance apart. The rationale of introducing the modified mutation operator (3) is that it is generally unlikely that the global maximum of J occurs at some narrow vicinity of the currently chosen reference point, but it is expected to be at some distance from it. Therefore, mutations generating should be most intensive some distance apart from the reference point, which is controlled by the parameter $s_{k,j}$.

Using the above mentioned ideas and their combinations (note that idea 2 should be technically combined with idea 1), we implemented the following optimisation algorithms:

- (M_0) Basic version of the algorithm.
- (M_1) Modification of the basic algorithm, where local optimisation search is added at points with $J(z_{k+1}) > \xi J_{max}$, $\xi \in [0,1]$ (Basic version + Idea 1)
- (M_2) Modification of the basic algorithm with the mutation kernel provided by the modified Gaussian distribution. (Basic version + Idea 3)
- (M_3) Modification of the basic algorithm, where local optimisation search is added at points with $J(z_{k+1}) > \xi J_{max}$, $\xi \in [0,1]$, and the length of the shift in the local optimisation is calculated based on the Lipschitz constant. (Basic version + Ideas 1,2)
- (M_4) Modification of the basic algorithm, where local optimisation search is added at points with $J(z_{k+1}) > \xi J_{max}$, $\xi \in [0,1]$, the length of the shift in the local optimisation is calculated based on the Lipschitz constant, and the mutation kernel is provided by the modified Gaussian distribution (Basic version + Ideas 1,2, and 3)
- (M_5) Modification of the basic algorithm, where mutation at points, where $J(z_{k+1}) > \xi J_{max}$, $\xi \in [0,1]$ is described by a normal distribution, whereas at points, where $J(z_{k+1}) < \xi J_{max}$, mutation is modelled by the modified Gaussian distribution (Basic version + Ideas 1,3)

3. Proof of convergence of the new optimisation algorithm

In this section, we rigorously prove the convergence of the basic optimisation algorithm introduced in Section 2.2. The proof is based on the two following theorems.

Theorem 1. *Consider the sequence σ_k given by $\sigma_k = 2R^2/\ln(k)$, where $R^2 = (a_1-b_1)^2+\dots+(a_m-b_m)^2$ (here a_i, b_i are real numbers). Then the sequence of point $\{z_k\}$, generated in the basic algorithm is everywhere dense in the box $\Pi = \{z = (x_1, x_2, \dots, x_m) : a_i < x_i < b_i, i = 1, \dots, m\}$ with the probability of unity. In other words, for any $z \in \Pi$ and any $\epsilon > 0$, the probability to have at least one point z_k in the neighbourhood $O_\epsilon(z)$ of z tends to unity when $k \to \infty$.*

Proof. Consider a positive number $\epsilon < 1/2$ and an arbitrary point $z = (x_1, x_2, \dots, x_m) \in \Pi$. We consider the following 'cubic' ϵ neighbourhood of z defined by $\omega = \{y = (y_1, \dots y_m) : |y_j - x_j| \le \epsilon, j = 1, \dots, m\}$. The probability $P_{k+1}(\omega)$ for the point z to land in ω is given by

$$P_{k+1}(\omega) = \frac{\int_\omega \exp\left(-(z_{k+1} - z'_{k+1})\Sigma_{k+1}^{-1}(z_{k+1} - z'_{k+1})^T\right) d\omega}{\int_\Pi \exp\left(-(z_{k+1} - z'_{k+1})\Sigma_{k+1}^{-1}(z_{k+1} - z'_{k+1})^T\right) d\Pi}.$$

By taking into account the fact that the diagonal elements $\sigma_{k+1,j}$ of the matrix Σ_{k+1} are either σ_{k+1} or σ_{k+1+M} and that $\sigma_k = 2R^2/\ln(k)$ (we remind that the integer $M > 1$ is a parameter of the optimisation method), the following inequalities will hold

$$1 > \exp\Big(-(z_{k+1} - z'_{k+1})\Sigma^{-1}_{k+1}(z_{k+1} - z'_{k+1})^T\Big) > \exp\Big(\frac{-\ln(k+M+1)|z-z'|^2}{2R^2}\Big)$$

$$= (k+M+1)^{\left(-\frac{|z-z'|^2}{2R^2}\right)} > (k+M+1)^{-\frac{1}{2}};$$

$$\int_{\Pi} \exp\Big(-(z_{k+1} - z'_{k+1})\Sigma^{-1}_{k+1}(z_{k+1} - z'_{k+1})^T\Big) d\Pi < \big(\pi\sigma_{k+1}\big)^{n/2} = \Big(\frac{2\pi R^2}{\ln(k+1)}\Big)^{n/2}.$$

We estimate the lower bound of the probability $P_{k+1}(\omega)$ as follows

$$P_{k+1}(\omega) \geq \Big(\frac{\ln(k+1)}{2\pi R^2}\Big)^{n/2}(k+M+1)^{-\frac{1}{2}}\prod_{j=1}^{m}\frac{2\epsilon}{(b_j - a_j)}.$$

We denote $\gamma = \big(2\pi R^2\big)^{-n/2}\prod_{j=1}^{m}\frac{2\epsilon}{(b_j-a_j)}$. This constant only depends on the size of the chosen neighbourhood but does not depend on the number of iterations in the algorithm.

The probability P_k to miss the ϵ- neighbourhood in each of k first iterations can be estimated as

$$P_k < (1 - \gamma(k+M)^{-\frac{1}{2}}(\ln 2)^{n/2})^k.$$

Since we have

$$\lim_{k\to\infty}(1 - \gamma(k+M)^{-\frac{1}{2}}(\ln 2)^{n/2})^k = 0,$$

then the probability of non-choosing a point from ω for k iterations should tend to zero with an increasing number of iterations.

Thus, we conclude that the sequence of points $\{z_k\}$ is everywhere dense in Π with the probability of unity. □

Theorem 2. *Consider a continuous positive function $J(z)$ defined in $\Pi = \{z = (x_1, x_2, \dots, x_m) : a_i < x_i < b_i, i = 1, \dots, m\}$, which has a unique point of maximum $z^* = (x_1^*, x_2^*, \dots, x_m^*)$. Consider the sequence σ_k, used in the optimisation method, to be $\sigma_k = 2R^2/\ln(k)$, where $R^2 = (a_1 - b_1)^2 + \dots + (a_m - b_m)^2$. Then for any $\epsilon > 0$, the probability of choosing a point z_k from the ϵ-neighbourhood $\omega(z^*)$ of the point z^* tends to unity when $k \to \infty$. Therefore, the sequence of points $\{z_k\}$ generated by the optimisation algorithm converges in probability to the point z^* of the global maximum of the function J.*

Proof. We estimate the probability for the point z_{k+1}, which is obtained at the $k+1$ iteration, to land in $\omega(z^*)$, i.e. the ϵ-neighbourhood of the point z^*: $\omega(z^*) = \{y = (y_1, y_2, \dots, y_m) : |y_j - x_j^*| \leq \epsilon, j = 1, \dots, m\}$. We also introduce the following notation: $J_0 = \sup_{z\in\Pi\setminus\omega(z^*)} J(z)$; I_k is the set of indexes i of the points $z_{k,i}; i = 1, \dots, n$ which land into $\omega(z^*)$; $\overline{I_k}$ is the set of indexes of the points $z_i, i = 1, \dots, k$ which land outside $\omega(z^*)$. According to Theorem 1, the sequence $\{z_k\}$ is everywhere dense (with the probability of unity) in Π. Since J is a continuous function, there will always be a point $z_p \in \omega(z^*)$ such that we have $J(z_p) > J_0$.

For $k > p$, the probability to choose the point z_j as the reference point from the set $\overline{I_k}$ can be estimated as

$$\frac{\Sigma_{j\in\overline{I_k}} J^{\phi_k}(z_{k,j})}{\Sigma_{j=1}^{n} J^{\phi_k}(z_{k,j})} \leq \frac{\Sigma_{j\in\overline{I_k}} J_0^{\phi_k} J^{-\phi_k}(z_p)}{\Sigma_{j=1}^{n} J^{\phi_k}(z_{k,j}) J^{-\phi_k}(z_p)} < n\Big(\frac{J_0}{J(z_p)}\Big)^{\phi_k} \xrightarrow{k\to\infty} 0$$

Since the above probability tends to zero, then the probability of choosing the point z_j as a reference point z' with the index j from the set I_k tends to unity. We estimate the probability $P_{k+1}(\omega(z^*))$ of choosing the point z_{k+1} of trial at $(k+1)$th iteration of the optimisation algorithm, under condition that the reference point z' for the given iteration is chosen from the set I_k, i.e. $z' \in \omega(z^*)$.

$$P_{k+1}(\omega(z^*)) = \prod_{j=1}^{m}\frac{\int_{x_j^*-\epsilon}^{x_j^*+\epsilon}\exp\big(-\frac{(y_j-x'_j)^2}{s_{k+1,j}}\big)dy_j}{\int_{a_j}^{b_j}\exp\big(-\frac{(y_j-x'_j)^2}{s_{k+1,j}}\big)dy_j}$$

$$\geq \prod_{j=1}^{m}\Big[\Phi\Big(\frac{x_j^* + \epsilon - x'_j}{s_{k+1,j}}\Big) - \Phi\Big(\frac{x_j^* - \epsilon - x'_j}{s_{k+1,j}}\Big)\Big],$$

where the function Φ is the Gauss error function:

$$\Phi(t) = \frac{2}{\sqrt{\pi}}\int_0^t \exp(-\tau^2)d\tau.$$

Since $z' \in \omega^*$, we have: $x_j^* - \epsilon < x'_n < x_j^* + \epsilon$, and $x_j^* + \epsilon - x'_j > 0$, $x_j^* - \epsilon - x'_j < 0$.

Since $1/s_{k+1,n} \to +\infty$ for $k \to +\infty$, then

$$\Phi\Big(\frac{x_j^* + \epsilon - x_j'}{s_{k+1,j}}\Big) \to 1, k \to +\infty; \Phi\Big(\frac{x_j^* - \epsilon - x_j'}{s_{k+1,j}}\Big) \to 0, k \to +\infty.$$

From the obtained above inequality for $P_{k+1}(\omega(z^*))$, it follows that $P_{k+1}(\omega(z^*)) \to 1$, for $k \to \infty$. This completes the proof of the theorem on the convergence of the proposed optimisation algorithm. □

4. Implementing the optimisation methods to multi-modal test functions

We apply the basic optimisation algorithm (M_0) as well as its modifications (M_i), $i = 1,2,3,4,5$ to various test functions $J(z)$. We also compare the results of optimisation with other algorithms of global optimisation. For comparison, we use the optimisation methods from the same class, i.e. they are both evolutionary and stochastic. These methods are also well-known in the literature to prove their efficiency. Here we consider the following methods: differential evolution, DE ('current to the best' algorithm) [33], ESCH (Evolutionary Strategy with Cauchy distribution) [30], CRS (Controlled Random Search with local mutation) [31,32], as well as the previously introduced modification SoFDE of the algorithm SoFA [26].

In our simulations, for each type of test function J, we run $N = 100$ optimisation trials. The sequence ϕ_k in the optimisation methods is given by $\phi_k = \sqrt{1 + k/10000}$. The parameter Q, describing mutations within the population, is set to be $Q = 0.1$. We set the total number of points n at each iteration (the population number) to be 100 in all simulations. The generic form of the test functions is given by $J(z) = 1/(1 + \exp(F(z)/m)) = J(F(z))$, where $F(z)$ are taken from the list of well-known functions, which are usually considered for testing global optimisation algorithms (artificial landscapes). The above mentioned transform from F to $J(z)$ is needed to guarantee the positivity of $J(z)$. We considered $F(z)$ to be the Rastrigin, Griewank, Levy, Styblinski–Tang, or GKLS multimodal functions [34,35] (the mathematical expression of each of the mentioned functions is provided below).

Rastrigin function. The function is described by the following equation:

$$F_1(z) = -Am + \sum_{i=1}^{m}\big(A\cos(2\pi x_i) - x_i^2\big), \tag{4}$$

where the dimensionality of space m is 50; $x_i \in [a_j, b_j]$, with the boundaries given by $a_j = -5.12$, $b_j = 5.12$, $A = 10$. The global minimum of the Rastrigin function is reached at the point $z = 0$, $F_1(0) = 0.5$. This corresponds to the global maximum of $J(z)$ being 0.5.

The outcome of implementation of the optimisation methods to the Rastrigin function is presented in Fig. 1A. This figure plots the maximal found value of $J(F(z))$ against the number of maximal iterations (i.e. the iteration number where an algorithm terminates).

Griewank function. The function is defined as follows:

$$F_2(z) = -\sum_{j=1}^{m}\frac{x_j^2}{4000} + 10\prod_{j=1}^{m}\cos\Big(\frac{x_j}{\sqrt{j}}\Big) - 10. \tag{5}$$

The function $F_2(z)$ is defined in the m-dimensional cube ($m = 50$), with $x_i \in [-100, 100]$, $i = 1, \dots, m$. The global minimum of the Griewank function is reached at the point $z = 0$, $F_2(0) = 0$. This corresponds to the global maximum of $J(z)$ being 0.5.

The performance of the considered optimisation methods applied to the Griewank function is presented in Fig. 1B.

Levy function. The function is defined as follows

$$F_3(z) = \sin^2(\pi w_1)^2 + \sum_{j=1}^{m-1}(w_1 - 1)^2\Big(1 + 10\sin^2(\pi w_j + 1)\Big) + (w_m - 1)^2\Big(1 + 10\sin^2(2\pi w_m)\Big), \tag{6}$$

where w_j are defined as $w_j = 1 + (x_j - 1)/4$, $j = 1, \dots, m$. We consider the function $F_3(z)$ in the m-dimensional cube ($m = 50$), with $x_j \in [-10, 10]$, $j = 1, \dots, m$. The global minimum of the Levy function is reached at the point $z = 0$, $F_3(0) = 0$. This corresponds to the global maximum of $J(z)$ being 0.5. The performance of the optimisation methods applied to the Levy function is shown in Fig. 1 C.

Styblinski–Tang function. The function is defined as follows

$$F_4(z) = \frac{1}{2}\sum_{j=1}^{m}\big(x_j^4 - 16x_j^2 + 5x\big) + Am, \tag{7}$$

where the constant A is determined from the condition that the global minimum of the function is zero ($A \approx 39.16599$). The function evaluated on the cube $x_j \in [-5, 5]$, $j = 1, \dots, m$. The point of the global minimum is given by $x_j \approx 2.903534$, $j = 1, \dots, m$. The global maximum for $J(z)$ is 0.5. The performance of the considered optimisation methods applied to the Styblinski–Tang function is presented in Fig. 1D.

From Fig. 1, one can see that the efficiency of an optimisation method can largely depend on the choice of a particular test function. On the other hand, in all considered test functions, the newly created method (M_2) seems to demonstrate the best performance. Furthermore, most of the other modifications show a better (or comparable) performance than the considered well-known heuristic algorithms DE, ESCH, and CRS. This demonstrates a great potential of considering anisotropic mutations in optimisation algorithms.

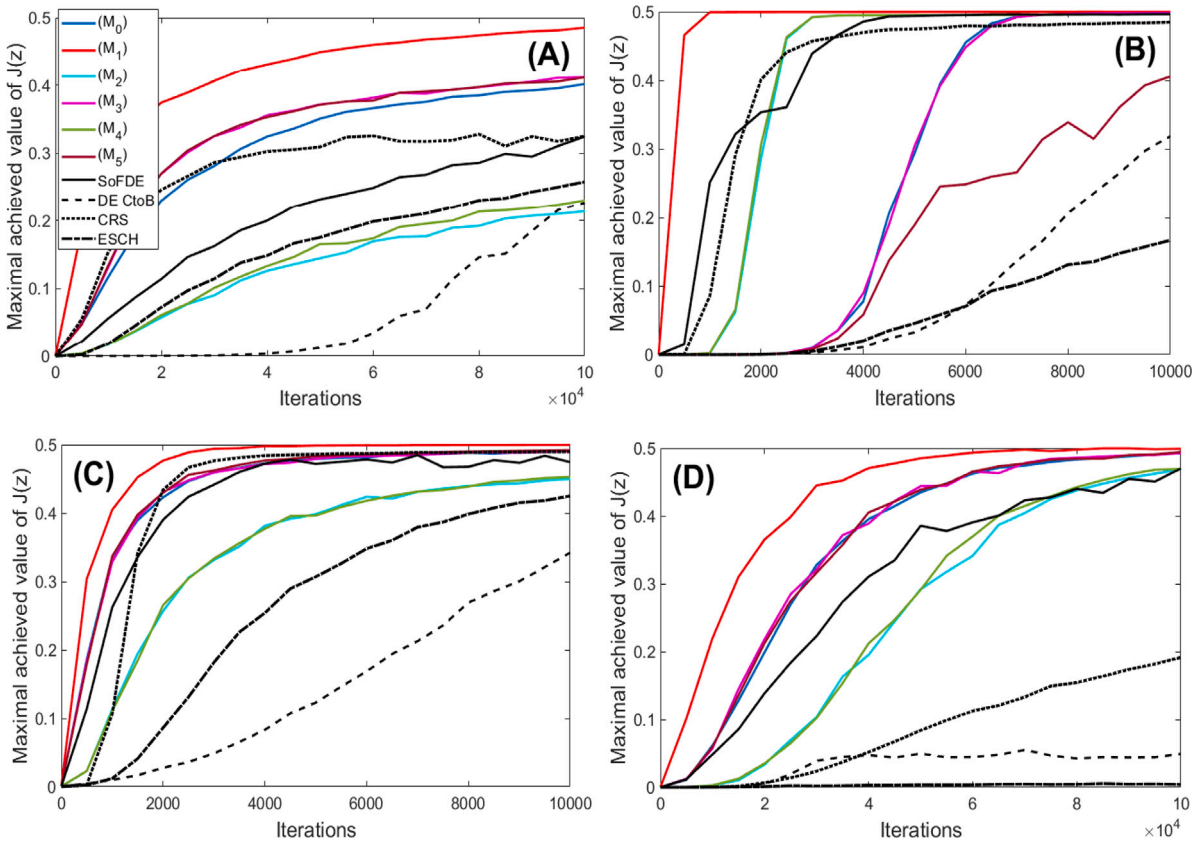

Fig. 1. Comparison of the efficiency of the optimisation algorithms applied to various test functions. We consider $J = 1/(1+\exp(F_i(z)/m))$, with different functions F_i, $i = 1, 2, 3, 4$. For each test function, the results are averaged over 100 realisation of the methods. The vertical axis shows the maximal value of the objective function achieved, the horizontal axis shows the maximal number of iterations. Panels (A)-(D) show the results for: (A) Rastrigin function (F_1); (B) Griewank function (F_2); (C) Levy function (F_3), and (D) Styblinski–Tang function (F_4). The mathematical description of the functions F_i is provided in the main text. The names of optimisation algorithms are labelled in panel (A) and it is valid of the other panels.

GKLS test functions. To reduce the influence of the choice of a particular test function on the comparison of performance of global optimisation algorithms, we further consider a broad class of non-differentiable stochastic test functions known as GKLS (ND-type). This class of test functions is characterised by a substantial flexibility of the position, the number and the height of the points of maximum. A detailed description of GKLS functions can be found in [36,37]. In our simulations, we randomly generated a test GKLS function with the number of extremum points to be 1000. The position of the global maximum was generated randomly. The optimisation was performed in the m-dimensional cube ($m = 50$), with $z_i \in [-10, 10]$, $i = 1, \ldots, m$. As in the previous examples, the global maximum was set to be always 0.5.

Fig. 2A shows the performance of the optimisation methods applied to GKLS test functions. For each optimisation method, we generated 100 samples of test functions. Therefore, for each iteration, each graph corresponds to the mean value of the $J(z)$, which is the average over 100 realisations of the test function.

We also compared the efficiency of the optimisation algorithms based on their ability to approach some fixed vicinity global maximum after a particular number of iterations. For the class of test GKLS functions, we generated 100 samples of non-differentiable GKLS functions with the number of extremum points to be 100. Each optimisation algorithm was tested on each randomly generated function for different number of interactions k. We apply the following metric of success of optimisation: the Euclidean distance between the true point of global maximum and its approximation should be smaller than some threshold value Δ_{cr}. In our simulations, we take $\Delta_{cr} = 0.1$. This procedure can be considered as a competition between the optimisation methods.

Fig. 2B shows the outcome of the competition between the global optimisation methods. One can see that the newly constructed algorithm (M_1) is the winners of such competition since it achieve 100% success in reaching the vicinity of maximum after smaller numbers of iterations as compared to DE, ESCH, CRS, and SoFDE. Other versions, such as (M_0), (M_3) and (M_5) provide comparable results with the existing stochastic algorithms. This demonstrates the potential applicability of newly constructed methods to solving a variety of optimisation problems.

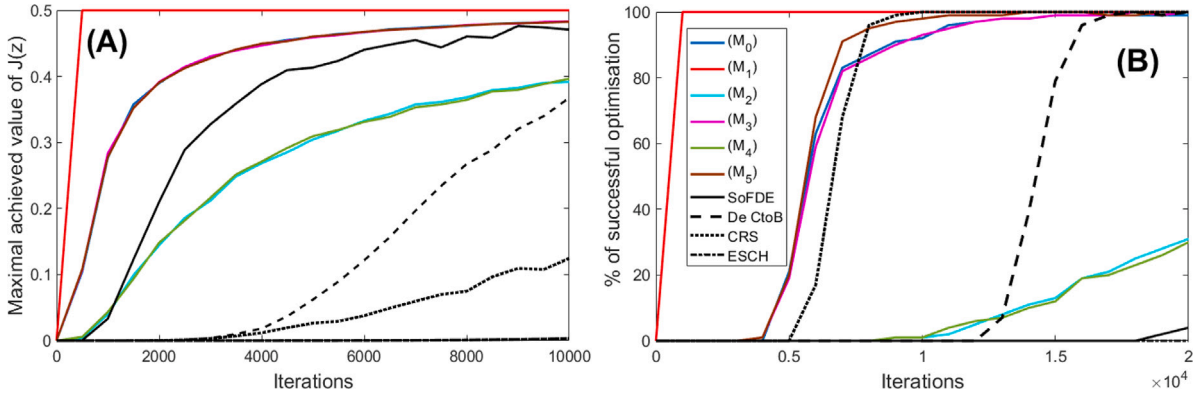

Fig. 2. Comparison of the efficiency of the optimisation algorithms applied to GKLS test functions $F(z)$. (A) The comparison metric is the maximal achievable value of $J(z)$. The vertical axis shows the maximal value of the objective function achieved, the horizontal axis shows the maximal number of iterations. (B) The comparison metric is the percentage of the successful implementations of the method with the condition being to land within Δ_{cr} neighbourhood of the point of maximum after the for various number of maximal iterations. The names of optimisation algorithms are listed in the figure labels. For details on numerical simulations see the text.

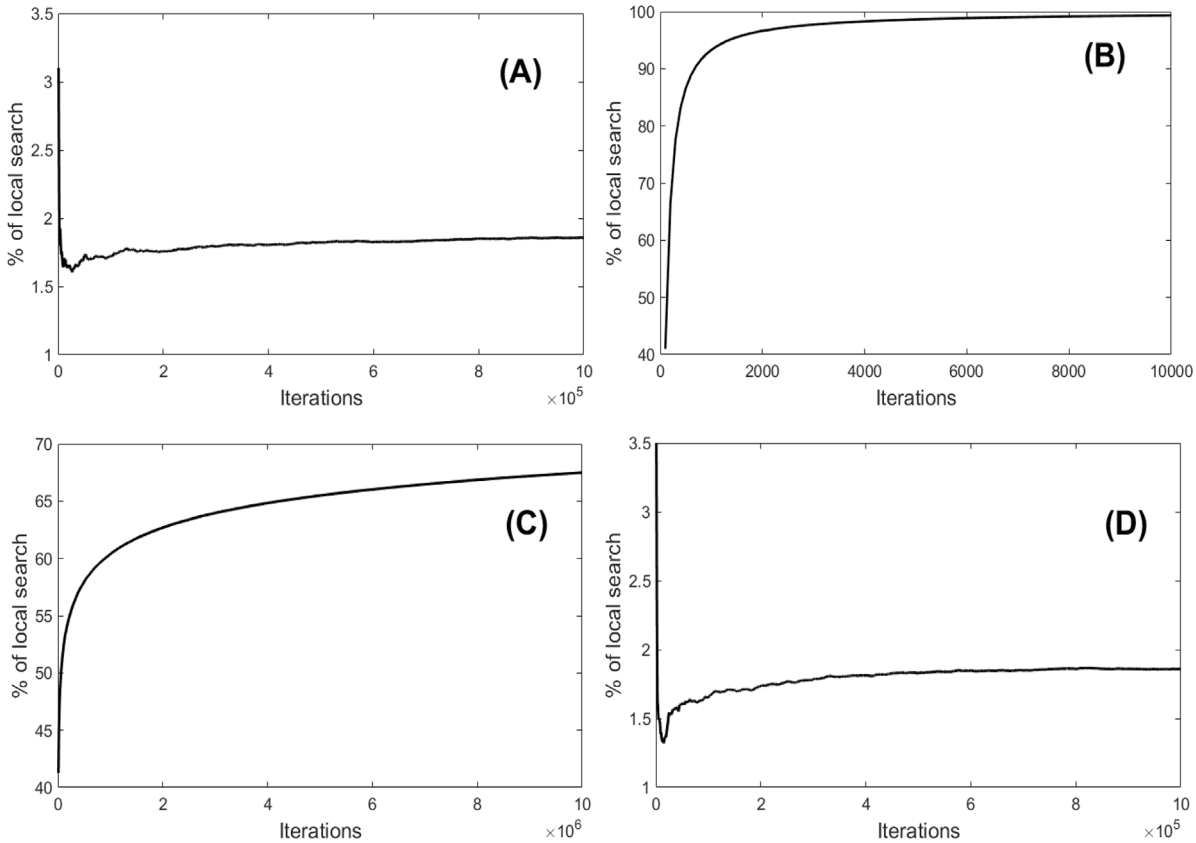

Fig. 3. Percentage of the local search (gradient-based) in the optimisation conducted using algorithm M_1. For each iteration number, the shown percentage is calculated as the number of realisations, where local search was applied, divided by the total number of realisations of the algorithm (100). Panels (A)-(D) correspond to optimisation of: (A) Rastrigin function (F_1); (B) Griewank function (F_2); (C) Levy function (F_3), and (D) Styblinski–Tang function (F_4).

Table 1
Various scenarios of the decay of the variance σ_k in the mutation kernel in optimisation algorithms (M_i) considered in Fig. 4.

Scenario number	Parameterisation of σ_k
1	$\frac{100}{\log_2(k+1)}$
2	$\frac{10000}{k+1}$
3	$\frac{100}{\log_2^2(k+1)}$
4	$\frac{10^2 \log_2^2(k+1)}{k+1}$
5	$\frac{250^2}{\log_2^4(k+1)}$
6	$\frac{100}{\sqrt{k+1}}$

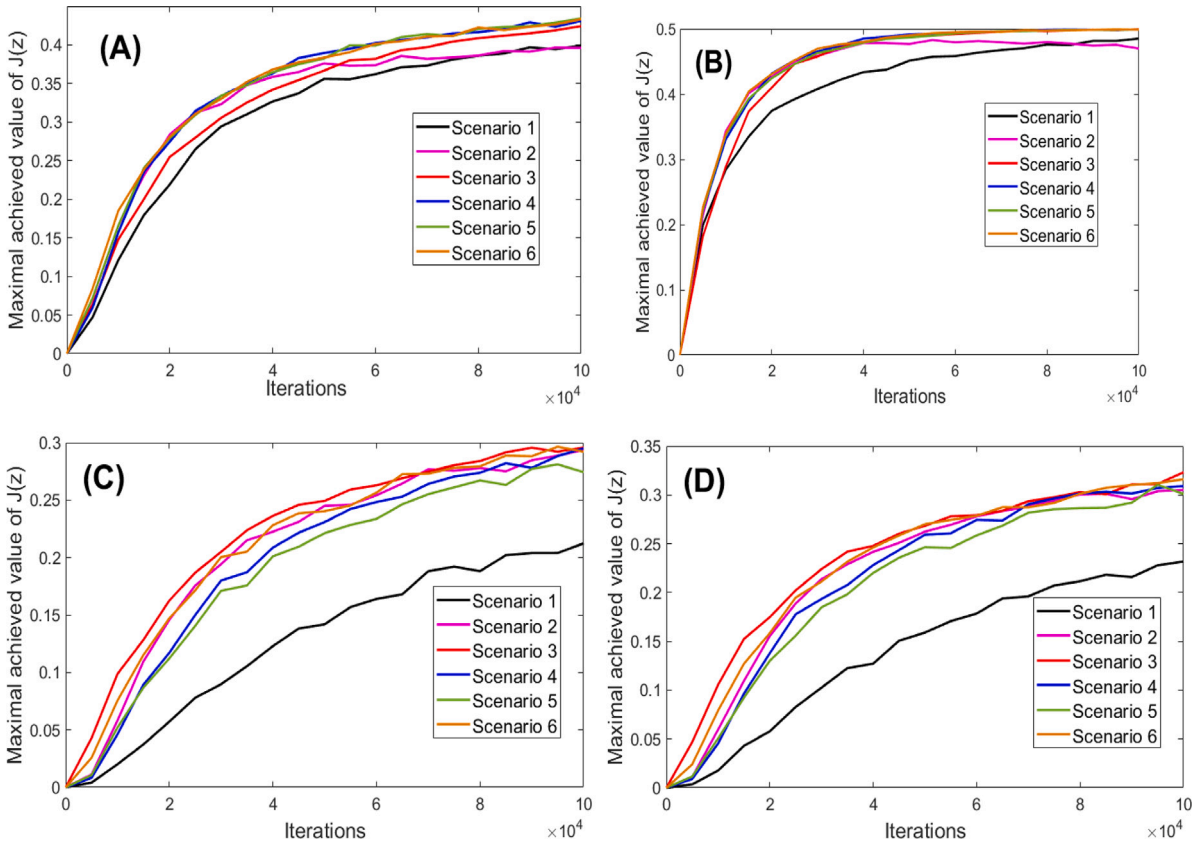

Fig. 4. Influence of the rate of decay of the variance of the mutation kernel (parameterised by $2\sigma_k$) on the convergence rate of the novel optimisation algorithms (M_i). The six presented scenarios (parametrisations) σ_k are listed in Table 1. The test function is the Rastrigin function (F_1). For each test function, the results are averaged over 50 realisation of the methods. In each panel, for a chosen optimisation algorithm (M_i). Panels (A)-(D) correspond to the following algorithms: (A) (M_0); (B) (M_1); (C) (M_2); (D) (M_4). The vertical axis shows the maximal value of the objective function achieved, the horizontal axis shows the maximal number of iterations.

From Figs. 1 and 2, one can see that out of the proposed novel algorithms (M_i), modification (M_1) seems to be the fastest one, and this highlights the importance of combining the local and the global search techniques. For (M_1), we conducted some further investigation to better understand the reason of its efficiency. In particular, we explored the relative frequency of implementing the local, gradient-based search, and the dynamics of this frequency as the number of iterations increases. The results are presented in Fig. 3, where the percentage of the local search (measured as the number of realisations using local search for the given iteration divided by the total number of realisations of the algorithm) is plotted against the number of iterations. We considered the four test functions as in Fig. 1. One can see that the fraction of the local search in the total search largely depends on the nature of the functions and it varies from low values (<2%) for the Rastrigin and Styblinski–Tang functions, to larger values of 70%–100% for

the Griewank and Levy functions. The graphs show an increase of the fraction of the local search with the number of iterations, however, for some functions (the Rastrigin and Styblinski–Tang functions), there can be a larger initial contribution of the local search. The latter effect can be explained by spending initially more time near some local maxima. Overall, Fig. 3 can shed some light on the better efficiency of (M_1) as compared to the basic algorithm (M_0). An adaptive switch between the local and the global search, depending the progress made in improving the objective function, seems to be the key point. For example, for large iterations, where we eventually arrive to the vicinity of the global maximum, focusing on local search can be counter-productive for some functions, as shown in Figs. 3A,D, whereas for other functions, local search in the vicinity of the global extremum is more efficient (Figs. 3B,C). We must stress, however, that a detailed research into the adaptive switch between local and global search in the proposed optimisation algorithms is still an open question and should be done elsewhere.

Finally, we find that the novel optimisation methods (M_i) can be adjusted in terms of their speed of approaching the optimal solution, by regulating the rate at which the variance in the mutation operator decreases with the iterations. In the previous simulations, we consider the sequence σ_k to decay as $1/\log_2(k)$, for which the convergence to the optimal solution is ensured by Theorem 2. Our simulations show, however, that a faster decay is possible for some other parameterisations of σ_k. The results of our simulations are presented in Table 1 and Fig. 4. In Fig. 4, we show how the six scenarios of decay of σ_k, listed in Table 1, modify the efficiency of the optimisation method applied the Rastrigin function in $m = 50$ dimensional space. Note that Scenario 1 corresponds to the results shown in Figs. 1, 2. For the sake of brevity, here we only show optimisation based on algorithms (M_0), (M_1), (M_2); and (M_4). One can see that generally, a proper tuning of σ_k can result in a faster optimisation for all methods, however, the effect is more pronounced for modifications (M_2), (M_4) and is less visible for the basic version (M_0) and (M_1). On the other hand, we should stress that the convergence of global optimisation has been rigorously proved only for the scenario $\sigma_k \propto 1/\ln(k)$ (scenario 1 in Table 1). Therefore the results for other parameterisations of σ_k should be considered, at least currently, as being heuristic: additional efforts are needed to rigorously prove their convergence on a broad class of functions. Importantly, we must say that one can always use a combined sequence σ_k, where the decay rate of mutations variance can be any function for some finite number N_1 of initial iterations, i.e. for $k = 0, 1, \dots, N_1$. Then, for all $k > N_1$, we can implement the parameterisation $\sigma_k = 1/\log_2(k)$, for which the convergence is guaranteed by Theorem 2.

5. Optimal control of disease spread using the new optimisation method

5.1. Mathematical approaches to control the spread of infections

An important practical application of optimisation methods is the optimal control of spread of infectious diseases with the main goal being reduction of severe human losses and economic cost [38–41]. In this study, we briefly show how our methods of global stochastic optimisation can be applied to find the optimal control strategy for infection spread within a heterogeneous population.

Currently, there exist few approaches to disease modelling. For example, the classical framework implements compartment-based modelling, where the system is described in terms of few possible states of hosts, for example, considering compartments of healthy, exposed, and infected hosts. The epidemic process is modelled as transition between those states. This approach assumes the homogeneity within the population, and operates with average characteristics of hosts (e.g. average rates of contacts and recovery) [42–45].

To take into account heterogeneity of hosts, multi-agent modelling framework has been created, assuming that the system consists of an ensemble of different interacting agents, described by distinct characteristics [46]. The behaviour of each agent in the model is defined using a set of simple rules. Note that by an agent, we can understand not only an individual, but a group of people with the same characteristics. Importantly, even for a small number of agents, the complexity of a multi-agent system increases dramatically, which requires substantial computational power to make predictions. A multi-agent system can be represented as a network, where edges describe interaction between the agents [47].

The specificity of a chosen disease modelling approach will strongly affect the way of finding the optimal control strategy. For example, in compartmental models, the optimal control of epidemics can be calculated using the classical Pontryagin's maximum principle [48], allowing for an analytical solution. However, the obtained optimal control solution will only operate with some average characteristics of the population. Therefore, the particularities of agents will be fully neglected. The optimal control in the multi-agent infection models will take into account individuality of each agents, thus making disease management more efficient. Infection management and control in a multi-agent system often consists in adaptive alteration of the strength of contacts between each pair of agents [49,50]. On the other hand, optimisation in multi-agent systems can be a very challenging task, due to a high dimensionality of the parameter space. Implementing evolutionary algorithms of global stochastic optimisation, is considered as a promising tool in solving problems of optimal control in high dimensional systems, as multi-agent epidemics models [51–53].

5.2. Model equations of spread of infections

In the example below, we use the multi-agent disease model adapted from [54]. We consider time-discrete settings, with the time increment being Δt (e.g. a day, a week, a month, etc.). We denote the time intervals, where model variables remain constant, by k, $k = 0, 1, 2, \dots, T$. We consider a population of N agents, which interact with each other. Contacts between healthy and infected agents may result in transmission of an infectious disease. The infection burden in the system is described in terms of the probabilities $p_{i,k}$ that the agent i stays infected within the time interval k. For simplicity, we assume that the probability of receiving infection by the agent i due to contacts with the agent j within the time period Δt is constant, and is determined by the parameter β_j. The probability

of recovering for the infected agent i within the period Δt is assumed to be constant as well, and is given by the parameter γ_i. We model the strength of contacts between the agent i and agent j within the time interval k by the parameters $w^0_{i,j,k}$, $0 \le w^0_{i,j,k} \le 1$, which have the meaning of weights.

The disease management of the system consists in adjusting the level of interactions/contacts between agents to impede disease transmission. Technically, this can be realised, for example, via applying some administrative restrictions, which can be different for different categories of agents (e.g. students, doctors, agricultural workers, etc.). The control of the system is, in general, time-dependent, i.e. it is different for different time intervals. Mathematically, this is described by alteration of the weights $w_{i,j,k}$ (to avoid confusion, the previously introduced weighs $w^0_{i,j,k}$ apply to the 'natural', i.e. without control, scenario).

The total probability of receiving infection by a healthy agent i within the time interval k can be obtained via summation of the weighted probabilities over all agents:

$$(1 - p_{i,k}) \sum_{j=1}^{N} w_{i,j,k} \beta_j p_{j,k}. \tag{8}$$

The multiplier $(1 - p_{i,k})$ describes the probability that the agent i was healthy in the beginning of the consider time interval k. The change in the probabilities $p_{i,k}$ over the time is governed by the following system of difference equations

$$p_{i,(k+1)} = p_{i,k} + (1 - p_{i,k}) \sum_{j=1}^{N} w_{ijk} \beta_j p_{j,k} - \gamma_i p_{j,k}, \tag{9}$$

where $i = 1, \dots, N$, $k = 1, \dots, T-1$. The initial state of the population, i.e. p_{i0}, $i = 1, \dots, N$, is assumed be known.

Using model (9), we can calculate the cost $f_i(p_{i,k})$ of infection burden experienced by the agent i within time interval k. The total cost of infection over the time of observation can be calculated by summing $f_i(p_{i,k})$ over the population and the time:

$$f = \sum_{k=1}^{T} \sum_{i=1}^{N} f(p_{j,k}). \tag{10}$$

In the case where we perform a control of the system, we need to add to expression (10) the cost of control of the system, which is defined as

$$\sum_{k=1}^{T} \sum_{j=1}^{N} \sum_{i=1}^{N} g_{i,j} \left(w_{i,j,k} - w^0_{i,j,k} \right), \tag{11}$$

where $w^0_{i,j,k}$ is the natural (i.e. without control) level of interactions between the agents; $g_{i,k}$ is the function describing the cost of limitation of contacts between the agent i and the agent j.

The optimisation task consists in finding the control strategy at each interval k (described by the set of weights $0 \le w_{i,j,k} \le 1$), such that the objective function f achieves its minimal value, under the constraint that the cost of realisation of control cannot exceed some level determined by a constant C:

$$g \equiv \sum_{k=1}^{T} \sum_{j=1}^{N} \sum_{i=1}^{N} g_{i,j} \left(w_{i,j,k} - w^0_{i,j,k} \right) - C \le 0, \tag{12}$$

Our goal is to provide an example of how the above optimal control problem can be solved using our global optimisation algorithms. Therefore, in this paper, we do not intend to preform any comprehensive investigation of influence of model parameters on the optimal control solutions: such analysis should be done elsewhere. We provide an illustrative numerical example of the optimal control for a single set of parameters (see next subsection). Finally, we should also stress that including constraint (12) into the optimisation problem was technically done using a penalty function (i.e. a penalty function was added to the objective function on the boundary of the valid domain).

5.3. Example of the optimal control of spread of infections

In this subsection we present an example of solving the optimal control problem for model (9) of the spread of infection. For simplicity, we consider the quadratic parameterisation of the functions f_i and $g_{i,j}$. The rationale of using the quadratic function is that at very low values of infection probabilities, the cost of infection for the society is negligibly small, whereas higher values of probabilities would signify substantial financial losses. The minimisation problem is formally stated as follows:

$$\begin{aligned} \min_{w \in W} \quad & \textstyle\sum_{k=1}^{T} \sum_{i=1}^{N} (p_{i,k})^2, \\ \text{s.t.} \quad & \textstyle\sum_{k=1}^{T} \sum_{j=1}^{N} \sum_{i=1}^{N} \left(w_{i,j,k} - w^0_{i,j,k} \right)^2 - C \le 0, \\ & 0 \le p_{i,k} \le 1, 0 \le w_{i,j,k} \le 1, \end{aligned} \tag{13}$$

where W is the set of all admissible weights $0 \le w_{i,j,k} \le 1$.

The above optimisation problem becomes highly multidimensional even for relatively small N, for example, in the considered case with $N = 7, T = 10$, the dimensionality of space is $m = N(N-1)T = 420$. To efficiently cope with a high dimensionality of the parameter space, we implement slightly modified versions of our optimisation algorithms. We introduce a modification of (M_0),

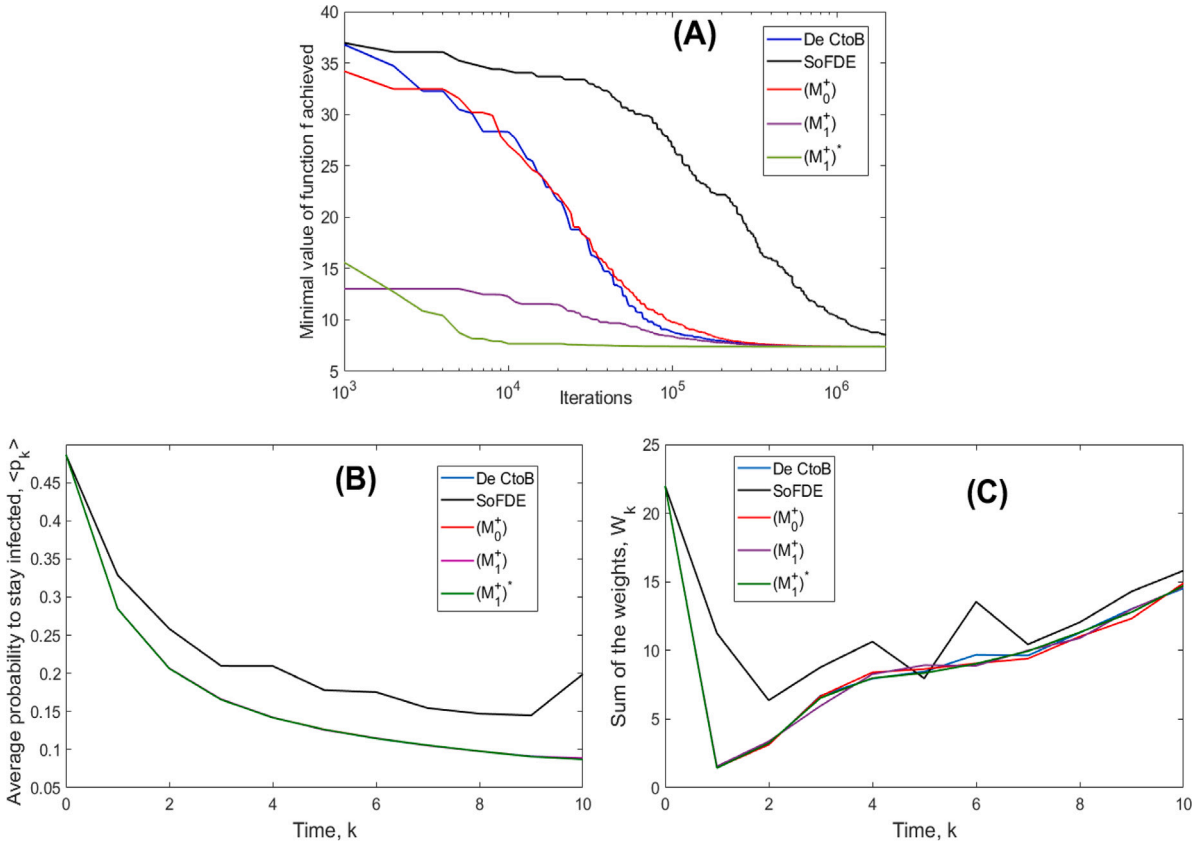

Fig. 5. Optimal control of the spread of infectious disease in model (9). The optimal problem is defined by (13). The details on the optimisation methods used are provided in the text. In methods (M_0^+), (M_1^+) and $(M_0^+)^*$ we used $\phi = k/10^{22}, \sigma = 10^{-8}/(\log_2(k+1))^2$; $Q = 0.7$; the variance of the weak mutation is given by 10^{-22}. (A) Minimal values of f achieved after performing a particular number of iterations. (B) The averaged over all agents probability $\langle p_k \rangle$ of staying infected within the time interval k defined by $\langle p_k \rangle = \sum_i p_{i,k}/N$. For each method, the values of $\langle p_k \rangle$ correspond to the best achieved solution. (C) The sum of weights W_k defined as $W_k = \sum_i \sum_j w_{i,j,k}$. For each method, the values of W_k correspond to the best achieved solution. The model parameters are $N = 7, T = 10$, $C = 0.2N(N-1)T = 84$. For simplicity $w^0 i, j, k$ are assumed to be time-independent ($w^0 i, j, k \equiv w^0 i, j$). The coefficients $\beta, \gamma, w^0 i, j$ as well as the initial conditions $p_{i,0}$ were taken as random variables uniformly distributed over the interval [0, 1].

further denoted by (M_0^+), where we use a new anisotropic mutation operator, which shifts the centre of mutation for all coordinates experiencing strong mutations (we remind that a strong mutation for coordinate i occurs in the case a randomly generated number within the interval [0,1] is smaller the parameter Q). For strong mutations, the new centre for the normal distribution of the mutation operator for the ith coordinate is given by: $x_i + h(X_i - x_i) + h(x_{ai} - x_{bi})$, where x_i is ith coordinate of the current reference point; X_i is the ith coordinate of the current point with the maximal fitness (the objective function); the points x_{ai}, x_{bi} are generated in the same way as the current reference point, $0 < h < 1$ (here $h = 0.8$). For all coordinates experiencing a weak mutation, the centre of the mutation operator is the coordinate of the current reference point.

We also introduce two other modifications of the optimisation algorithms, denoted by (M_1^+) and $(M_1^+)^*$, where we use the same principles as in M_0^+, but we add the local, gradient-based search, as in (M_1). The main difference between (M_1^+) and $(M_1^+)^*$ is regarding the number of local steps made. In (M_1^+), we use a fixed number of local steps (their length is given by 0.125; 0.25; 0.5; 1; 2; 3; 4; 8; 12; 16) to check the improvement, whereas for $(M_1^+)^*$, the number of local steps is not fixed but, the length of steps is determined by the sequence $2^n \cdot 10^{-11}$: we increase n until improvement of optimisation ceases.

Fig. 5 shows the performance of optimisation algorithms in finding the optimal control solution. For comparison, we also present the results obtained using the DE algorithm (current to the best, DE_CtoB) and the SoFDE algorithm. Fig. 5A depicts the evolution of the minimal value of the cost function f achieved for increasing numbers of iterations. Fig. 5B presents the decrease of the infection burden (infection level) with time (measured as $\langle p_k \rangle = \sum_i p_{i,k}/N$), corresponding to the best achieved solution. Fig. 5C shows the total sum of weights ($W_k = \sum_i \sum_j w_{i,j,k}$), corresponding to the best achieved disease management. One can see that the novel optimisation algorithms demonstrate a high rate of convergence, which is either better or comparable with the previous methods. From the disease management perspective, one can see from Fig. 5C that the best strategy consists in an immediate reduction of connectivity between the agents in the beginning of control (i.e. to have low W_k), with a further gradual increase of connectivity between agents at later times, where the infection burden becomes lower.

6. Discussion

Methods of global stochastic optimisation, in particular, evolutionary algorithms, are widely applied in modelling in physics, economics, finance, biology, and in minimising effects of natural disasters, such as epidemics. Despite the astonishing abundance of various evolutionary algorithms of different kinds [5–7], the great majority of them are *heuristic*: for them the fact of convergence of solution to the global extremum for an arbitrary multi-modal function is still to be rigorously proved. Clearly, this situation is unfortunate since application of such heuristic algorithms hinges on our faith and/or on our previous positive experience, rather than on some mathematical fact. For example, implementation of different optimisation to the same problem may provide different 'optimal' solutions (see examples in [21]).

For the approach proposed here, we are able to present a rigorous proof of convergence, which was a serious challenge for previous methods of global stochastic optimisation. Our algorithm is inspired by the survival on the fittest principle in evolutionary biology established by Charles Darwin. We consider points in the parameter space as interacting individuals (genotypes), which compete with each other for some limited resource. Under this scenario, achieving the point of global maximum in the space of life history traits will be the eventual outcome of natural selection of mutating genotypes. We must say that the principle of the survival of the fittest was considered in few previous studies, for example, in realisation of the methods SoFA and SoFDE [24,26]. However, the speed of the original algorithms, using the survival on the fittest principle, was somewhat slow, which prevented their wide practical usage. Here we argue that allowing for anisotropic mutations in the space of parameters and also combining the local and the global search for the optimal solution would largely improve the performance of the survival of the fittest algorithms and would make them highly competitive with the existing evolutionary algorithms. Overall, incorporating anisotropic mutations seems to be a new and promising idea in evolutionary optimisation, which has not be well exploited yet.

6.1. Implementing mutation and selection in evolutionary algorithms

In general, algorithms of evolutionary optimisation incorporate two pivotal processes which underline biological evolution. These are: (i) updating the genetic code due to permanent mutations, and (ii) selection of the fittest genotypes due to competition for resources [9,10,55]. Generation of mutations in optimisation is required to be able to scan a particular domain in the parameter space, where the point of the global maximum (the maximal fitness) is likely to be located, whereas selection is needed to reveal such a domain. The resultant random search for the optimal solution can be considered as some variety of the classical Monte-Carlo method, where the trial points are distributed non-uniformly: at each iteration the distribution of trial points depends on the configuration of the performance achieved at previous iterations. We argue that for efficient optimisation, we need a balanced incorporation of the both processes. For example, excessive generation of mutations would result in degenerating of the proposed algorithm to the classical, but largely inefficient, Monte Carlo method. On the other hand, prevailing selection over mutations would lead to a local optimisation with a possibility to miss the global maximum.

In the existing evolutionary algorithms, the implemented selection operator is often a simple one: it is based on a direct comparison of the value of the objective function (evolutionary fitness) for a pair of points, which imitate genotypes. Based on such pair-wise comparison, we choose the point, for which fitness is higher [15,17,18]. We argue that in biology selection should be a more complex process: a larger fitness manifests itself in better potential for producing offspring and this potential should be evaluated across the entire population rather than in a pair-wise way. Therefore, a major difference between the presented here framework and the classical evolutionary algorithms is that the discrepancy in fitness across the *total* population determines the chance of successful reproduction (item 1 in each iteration of our method). This approach follows the classical paradigm of in ecology [56].

In the literature, genetic variability is usually mimicked by two operators: mutations and recombination (crossover) [8–11]. Mutation operators allow to generate new (offspring) elements in the parameter spaces, which can occur within a wide neighbourhood of the parent element (point), which provides a basis for scanning the search area. The crossover operator promotes effects of genetic variability, by making changes in only a part of coordinates (genetic code) of the parent and keeping the other coordinates intact. This process ensures a larger scattering of genetic mixing in the space of parameters around the parent element. It was previously shown that the crossover is a key element in evolutionary algorithms since without it, the optimisation speed substantially drops [15,17,18].

The previous version of the method 'Survival of the Fittest Algorithm' (SoFA) utilised the operator of isotropic mutations in all directions in the search space. This operator was an analogue of the well-known operators of mutations, which have been widely used in previous evolutionary algorithms. However, the SoFA approach lacked a crossover operator, which negatively affected its efficiency. On the other hand, mechanistically adding a crossover operator into the SoFA cannot be a solution of the problem, since this would create problems with a rigorous justification of the formal convergence of the modified algorithm, which was proved for the original version of SoFA. In the algorithms proposed here, the mentioned difficulty is overcome by allowing mutations to be anisotropic. Technically, mutation along different coordinates in the space of parameters is described by an stochastic operator with a randomly oscillating variance. Arguably, the operator of anisotropic mutation with randomly chosen variance can be considered as a combination between mutation and recombination (crossover). We should stress that variability of mutations in different direction is purely random, i.e. non-adaptive. Our test simulations in high-dimensional space showed that random modification of mutation in different directions results in a substantial gain in the performance of the algorithm, for example with the algorithm SoFDE. Importantly, the fact of convergence of the method with probability of one can be rigorously demonstrated for anisotropic mutations, whereas it is hard to prove for the case of a pure crossover. Finally, we should admit that the exact mechanism of how the proposed random non-adaptive variation of mutation kernels results in a substantial progress in optimisation as compared to the case of isotropic mutations is still to be investigated.

6.2. Global optimisation as a process of extracting information about the system

In optimisation algorithms using iterations, a major problem is to specify after which iteration we can terminate the process to be able to achieve the required accuracy. This is usually a non-trivial problem since in most of practical applications, we do not know *a priori* the exact optimal solution, as for example, in the above considered multi-agent disease model. The simplest criterion of termination can be setting a constraint on the maximal number of iterations, and this approach is very popular in various numerical methods of optimisation [57,58]. The major drawback of this approach is that there is no guarantee that after a fixed number of iterations the required accuracy would be reached. Another well-known criterion of termination is that after having completed a number of iterations, the solution exhibits only some negligible improvement. However, for heuristic optimisation algorithms, there is a danger that the apparent lack of improvement of optimisation can be because of being trapped in the vicinity of some local extremum, which will result in a wrong conclusion.

A different approach to define termination in some evolutionary or swarm optimisation algorithms is based on the reported phenomenon that the search points have a trend to aggregate within a narrow domain after a number of iterations: such a domain is likely to be a point of global extremum [10–12]. However, we should stress that the mentioned effect of the concentration of points is mostly demonstrated in simple test functions, to illustrate the convergence of the method in principle, but only rarely in complex practical applications. This discrepancy is due to technical complications of evaluating of the concentration of the search points in a particular neighbourhood. For algorithms, which convergence is not proved rigorously, we have again the same danger of getting stuck in the vicinity of a false solution — a local extremum. On the other hand, the proposed algorithms suggest a fully different approach to determining criteria of termination. This is related to the phenomenon of the localisation of measure and concentration of search points, which we will discuss below [22].

We should notice that the effect of concentration of points in the parameter space is closely related to the process of receiving information about some unknown system under investigation. Indeed, generally, information can be understood in the classical sense of the Shannon–Hartley theory as the process of selecting 'small' subset out of some larger initial set [59]. For a rigorous mathematical formalisation of extracting information, one can implement the same mathematical philosophy, as it was developed to formalise the selection processes [22,60]. Namely, within a large set of elements, we introduce the indicator of presence measuring the degree of belonging of elements to some selected subset. Often, such an indicator can be considered to be the probability that a element belongs to the selected subset. This idea is similar to the mathematical concept of fuzzy sets [61]. The amount of information is quantified by the degree of heterogeneity of the distribution of the introduced indicator. Note that this approach is in a good agreement with the current understanding information as a pattern or a distribution [62]. As a result of selection (e.g. natural selection), there is a concentration of the indicator of presence over some selected subset. We argue that global stochastic optimisation uses similar philosophy. Indeed, the main aim of optimisation is obtaining the information about the location of the point of the global extremum, i.e. extracting a sufficiently narrow vicinity of this point. Therefore, initially, under lack of accurate information, the indicator of presence is distributed homogeneously across the whole space of search, since the point of global extremum can be potentially everywhere. Further, as we process the results of trials, the distribution of the indicator of presence becomes more heterogeneous. Eventually, for a correct solution of the optimisation problem, the indicator of presence is concerted in the desired subset containing the global extremum.

We should stress that the mentioned links between optimisation, information and selection processes are common for many methods of global stochastic optimisation; however, they especially manifest themselves in the survival of the fittest approach. In this case, the indicator of presence can be taken as the probability of choosing a new point z_{k+1} at each iteration. From the theorems in Section 3, it follows that as far as iterations progress, there the indicator of presence will accumulate in the vicinity of the point of the global maximum. Since the probability of choosing point z_{k+1} is updated at each iteration, we can now rigorously formulate the corresponding criterion of termination of the algorithm. Namely, we can stop our iterations in the case where the probability to find the point z_{k+1} in the vicinity of the currently chosen point z_k becomes higher than a certain fixed threshold value. This threshold value will measure the concentration of points. The proposed criterion is illustrated in Fig. 6, where we show evolution of the probability density functions (pdf) of generating the next point z_{k+1} with an increase in the number of iterations. In the simplest one-dimensional case ($m = 1$), the probability density function for generating the point z_{k+1} is defined as

$$pdf = \sum_{i=1}^{n} \frac{J^{\phi_k}(z_{k,i})}{J^{\phi_k}(z_{k,1}) + J^{\phi_k}(z_{k,2}) + \cdots + J^{\phi_k}(z_{k,n})} A_{k+1,i} \exp\left(-\frac{(z_{k+1} - z_{k,i})^2}{\sigma_{k+1}}\right), \tag{14}$$

where $A_{k+1,i}$ are the normalisation constants defined in Section 2.2.

In Fig. 6, we show application of the basic optimisation algorithm (M_0) to the one-dimensional version of the Rastrigin function, i.e. $J(z) = J(F_1(z))$. From the both panels of this Figure, one can see that for small number of iterations, the pdf shape is multi-modal. However, with an increase of iterations, the pdf distribution becomes uni-modal and is concentrated around the point of the global maximum $z = 0$, the shape of this distribution tends to a delta function. In this case, the termination criterion can be the requirement that the width of the resultant pdf should be smaller than some specified accuracy threshold.

6.3. Final remarks and future directions

As modifications of the basic algorithm (M_0), we considered hybrid versions, where global optimisation is combined with local (gradient-based) search. The underlying idea is somewhat similar (although not being equivalent) to splitting the population of searchers into explorers (scouts) and exploiters previously proposed in swarm optimisation [7,11,12]. We demonstrate that adaptive

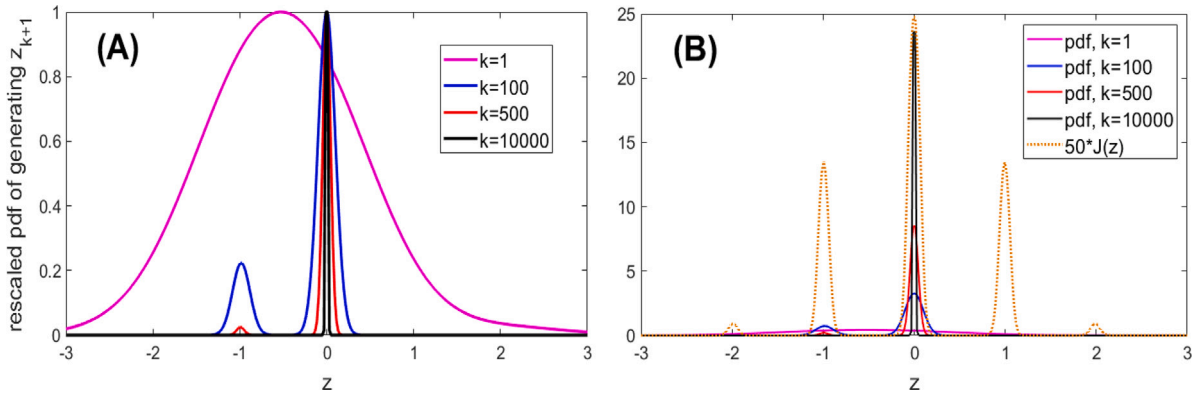

Fig. 6. Evolution of the probability density function (pdf), describing generation of the next point of search z_{k+1}, when the number of iterations k is being increased. The definition of the pdf is given by Eq. (14). Here the algorithm (M_0) is applied to maximisation of the one-dimensional fitness using the Rastrigin function $J(z) = 1/(1+\exp(F_1(z)/m))$. (A) The graphs for pdf of generating z_{k+1} are re-scaled to have one as the their maximal values. (B) The original (non-rescaled) graphs of pdf are presented along with the fitness function $J(z)$, which is multiplied by the factor of 50 for visibility. One can see a gradual concentration of the probability around the point of the global maximum at $z = 0$ as a result of optimisation.

switch between local and global approaches of can be particularly efficient: on the given test multi-modal functions, the algorithm (M_1) outperforms all other algorithms. Interestingly, the fraction of local exploration in the overall search can be as small as 1.8% or even less, but this would ensure the superiority of the algorithm over its basic version (M_0), utilising global search only. The switch between the local and the global search largely depends on the nature of the objective function $J(z)$ (see Fig. 3). On the other hand, the choice of a particular way to perform local optimisation under global–local hybrid schemes is be crucial. Indeed, implementing a different method for the local optimisation part, one can slow down the speed of convergence, for example, the method (M_3), using estimates of the local Lipschitz constant, has a somewhat inferior performance than compared to (M_1). Combining the global and the local optimisation showed its efficiency in optimisation in the multi-agent infection spread model (see Section 5, where we introduced modifications (M_1^+) and $(M_1^+)^*$ of (M_1)). Overall, a proper research into the combination of global and the local optimisation seems to be an interesting direction for future research.

The proposed methods seem to be flexible in terms of accelerating the speed of search by adjusting the decay rate of the variance of the mutation kernel σ_k. Initially, we investigated 6 different scenarios of parmeterisation of σ_k listed in Table 1, this demonstrated a potential for improving performance for most of the algorithms. We argue that a further research should be done into revealing the best rate of decay of the mutation variance. On the other hand, we should admit that so far, we manage to rigorously prove the convergence to the optimal solution for only one type of parameterisation, $1/log(k)$. A straightforward application of the same theoretical techniques to prove convergence for non-logarithmic functions σ_k seems to be hard, therefore, one will need to develop some different approaches.

Finally, the proposed optimisation methods are originally designed to be applied to a finite dimensional space. However, they can be easily adapted to work in functional (Hilbert) spaces, where space dimension is formally infinitely large. The modified basic algorithm of optimisation for Hilbert spaces is provided in Appendix A. As for the case, of optimisation in finite dimensional spaces, one can rigorously prove the convergence of the method in Hilbert spaces for a continuous positive function(al). The proof is somewhat similar to the one presented in Section 3 and is presented in Appendix A. We should stress that a further study would be needed to assess the rate of convergence and efficiency of the adaptation of the evolutionary algorithm to Hilbert spaces. This might contribute to the long-standing debates on the curse or bless of dimensionality in optimisation in very high dimensional spaces [63,64].

7. Summary

Two major factors are crucial for evolutionary methods of global optimisation: (i) the fact of convergence of a method when applying it to a broad class of objective functions, and (ii) the rate of achieving the point of global extremum, i.e. how fast the method is. Most of the existing evolutionary methods of optimisation show suitable performance in terms of speed, however, they remain heuristic since there is no guarantee that the optimal solution will be eventually achieved for any objective function within some broad class. In this study, we propose several optimisation algorithms, for which the fact of convergence (with probability one) can be rigorously proved. The methods are based on the famous principle of the survival of the fittest in evolutionary biology. The particularity of the novel optimisation methods is that they incorporate the idea of anisotropic mutation in the space of parameters, which has not been widely implemented in any of previous algorithms so far. Application of the constructed methods to a number of test functions in a highly dimensional parameter space (e.g. $m = 50$) demonstrates their better performance than compared to some other stochastic algorithms of global optimisation. Our algorithms showed themselves to be efficient in finding the optimal control solution in a high-dimensional multi-agent disease model.

CRediT authorship contribution statement

Oleg A. Kuzenkov: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Andrew Yu. Morozov:** Formal analysis, Methodology, Supervision, Investigation, Writing – original draft, Writing – review & editing. **Samvel A. Nalchajyan:** Investigation, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

The authors thank two anonymous reviewers for their commets which largely helped us to improve the manuscript.

Appendix A

Here, we adapt the Basic Algorithm of optimisation from the main text to solving problems in infinite dimensional (functional) spaces. We denote this algorithm as (M_{inf}). Firstly, we formalise the global optimisation problem in Hilbert spaces. Then, we describe the optimisation algorithm. Finally, we provide a rigorous proof of convergence of the algorithm for optimisation in infinite dimensional spaces.

Problem statement

We consider a Hilbert space l_2 of functions, which are described by summable sequences of numbers z, i.e.

$$l_2 = \left\{ z : z = (x_1, x_2, \dots, x_i, \dots), \left(\sum_{i=1}^{\infty} x_i^2 \right)^{1/2} < \infty \right\}.$$

In the above space, we choose an element $a = (a_1, a_2, \dots, a_i, \dots)$ with strictly positive components. We denote $R = \sqrt{\sum_{i=1}^{\infty} a_i^2}$. Using the element a, we introduce the following infinite dimensional cube given by $\Pi' = \{z : |x_i| \le a_i/2, i = 1, 2, \dots\}$. Let $J(z)$ be a continuous positive function defined in Π'. We assume that the function $J(z)$ has a unique point of global maximum (denoted by z^*) in Π'. The optimisation problem consists in finding the point z^*.

Description of the basic algorithm of optimisation (M_{inf})

As for the finite dimensional optimisation, the algorithm uses two sequences: ϕ_k and σ_k, for which we require $\lim_{k\to\infty} \phi_k = \infty$ and $\lim_{k\to\infty} \sigma_k = 0$. The other parameters of the method are: a real number $Q \in [0, 1]$, and a natural number $M > 1$.

Firstly, we initialise the method (considering this as a 'zero' iteration, $k = 0$), by generating a uniformly distributed set of points $z_{01}, z_{02}, \dots, z_{0n}$ in the box Π'. Here $z_{i0} = (x_{01i}, x_{02i}, \dots, x_{0ni}, 0, 0, \dots)$. Each coordinate is randomly chosen out of the segment $[-a_j/2, a_j/2]$, $j = 1, 2, \dots, n$. We evaluate the function $J(z)$ at each point: $J(z_{01}), J(z_{02}), \dots, J(z_{0n})$.

Next, at each iteration $k + 1$ $(k > n)$, we perform the following operations:

1. (Competition) Choosing the reference point. We select a reference point $z' = (x'_1, x'_2, \dots, x'_k, 0, 0, \dots)$ out of the current points $z_{k1}, \dots, z_{kn}$ with the probability defined by the ratio $J^{\phi_k}(z_{ki})/(J^{\phi_k}(z_{k1}) + J^{\phi_k}(z_{k2}) + \dots + J^{\phi_k}(z_{kn}))$, where $i = 1, \dots, n$. The chosen reference point is used in the further procedure.
2. (Mutation) Using the reference point z', we choose the point $z_{k+1} = (x_{k+1,1}, \dots, x_{k+1,k+1}, 0, 0, \dots)$ for a further test. The jth component of this point $(j = 1, \dots, k + 1)$ is randomly chosen from the interval $[a_j, a_j/2]$ with the probability defined by the normal law with the probability density function $A_{k+1,j} \exp(-(x_j - x'_j)^2/(s_{k+1,j}))$. Here, $s_{k+1,j}$ is the variance of the considered normal distribution divided by 2. We set $s_{k+1,j}$ as follows. We choose a uniformly distributed random number q_j from the interval $[0, 1]$, and we compare q_j with the parameter Q: in the case $q_j < Q$, then $s_{k+1,j} = \sigma_{k+1}$, otherwise, we set $s_{k+1,j} = \sigma_{k+1+M}$. The constants $A_{k+1,j}$ are chosen to normalise the probabilities, so integration of the probability density function over $[a_j, b_j]$ should give 1. We highlight that at each iteration, the number of non-zero components of the trial point increases by one.
3. (Selection) The function J is evaluated at the newly found point z_{k+1}. In the case, where $J(z_{k+1}) > J(z')$, the reference point is replaced by z_{k+1}, and the population of n points becomes updated.

The method will terminate after having achieved either of the following conditions: (a) the maximal possible number of iterations is achieved, or (b) the distribution of generated points around the one with the maximal J approaches the δ-function (see Discussion for more detail).

The proof of the convergence of the above optimisation algorithm designed for Hilbert spaces is provided below.

Proof of convergence of (M_{inf}) in Hilbert spaces

Theorem 3. *Consider the sequence σ_k given by $\sigma_k = 2R^2/\ln(k)$, where $R = \sqrt{\sum_{n=1}^{\infty} a_n^2}$. Then the sequence of point $\{z_k\}$, generated in the basic algorithm is everywhere dense in the infinite dimensional cube $\Pi' = \{z : |x_i| \le a_i/2, i = 1, 2, \dots\}$ with the probability of unity.*

Proof. Consider a positive number $\epsilon < 1/2$ and an arbitrary point $z = (x_1, x_2, \dots., x_i, \dots) \in \Pi'$. We consider the ϵ neighbourhood of z denoted as $O_\epsilon(z)$. Since the series $\sum_{i=1}^{\infty} a_i^2$ is convergent, there exist a number N such as $\sum_{i=N+1}^{\infty} a_i^2 \le \epsilon^2/2$. We take an arbitrary point $y = (y_1, \dots, y_i, \dots) \in \Pi'$ such that $|y_i - x_i| \le \epsilon/2^i$ for $i = 1, 2, \dots, N$. In this case, the following inequalities should hold

$$\sum_{i=1}^{\infty}(x_i - y_i)^2 = \sum_{i=1}^{N}(x_i - y_i)^2 + \sum_{i=N+1}^{\infty}(x_i - y_i)^2 < \sum_{i=1}^{\infty}\left(\frac{\epsilon}{2^i}\right)^2 + \sum_{i=N+1}^{\infty}(a_i)^2 < \epsilon^2$$

This signifies that $y \in O_\epsilon(z)$. Therefore, the set defined as $\omega = \{y = (y_1, \dots, y_i, \dots) : |y_i - x_i| \le \epsilon/(2^i), i = 1, \dots, N\}$ is a subset of the neighbourhood $O_\epsilon(z)$.

Assume that the kth iteration is completed, $(k > N)$. The probability $P_{k+1}(\omega)$ for the point z_{k+1} to land in ω can be expressed as

$$P_{k+1}(\omega) = \prod_{i=1}^{N} \frac{\int_{x_i-\epsilon/(2^i)}^{x_i+\epsilon/(2^i)} \exp\left(-\frac{(y_i - x_i')^2}{s_{k+1,i}}\right) dy_i}{\int_{-a_i/2}^{a_i/2} \exp\left(-\frac{(y_i - x_i')^2}{s_{k+1,i}}\right) dy_i}$$

Next we take into account the fact that the elements $s_{k+1,i}$ are either σ_{k+1} or σ_{k+M+1} (the integer $M > 1$ is a parameter of the optimisation method). Moreover, we recall the restrictions $|y_i - x_i'| < a_i$, $\sum_{i=1}^{N}(a_i)^2 < R^2$, and the equality $\sigma_k = 2R^2/\ln(k)$. The following inequality holds:

$$\int_{-a_i/2}^{a_i/2} \exp\left(-\frac{(y_i - x_i')^2}{s_{k+1,i}}\right) dy_i < \left(\pi\sigma_{k+1}\right)^{1/2} < \left(2\pi R^2/\ln(k+1)\right)^{1/2}.$$

Then we can evaluate the probability $P_{k+1}(\omega)$ as follows:

$$P_{k+1}(\omega) > \prod_{i=1}^{N} \frac{\epsilon\sqrt{\ln(k+1)}}{\sqrt{2\pi R^2}a_i 2^i} \exp\left(-\frac{\ln(k+M+1)a_i^2}{2R^2}\right) > \frac{(\ln(k+1))^{N/2}}{\sqrt{k+M+1}} \prod_{i=1}^{N} \frac{\epsilon}{R\sqrt{2\pi}a_i 2^i} = \frac{(\ln(k+1))^{N/2}}{\sqrt{k+M+1}}\gamma,$$

where we denote $\gamma = \prod_{i=1}^{N} \frac{\epsilon}{R\sqrt{2\pi}a_i 2^i}$. It is easy to see that γ only depends on the size of the chosen neighbourhood, but does not depend on the number of iterations of the algorithm.

The probability P_k' of missing the set ω by all points, which are generated within k iterations: from iteration $N+1$ to iteration $N+k+1$, can be estimated for large k as follows

$$P_k' < \left(1 - \frac{(\ln(N+1))^{N/2}}{\sqrt{k+M+1}}\gamma\right)^k.$$

However, since

$$\lim_{k\to\infty}\left(1 - \frac{(\ln(N+1))^{N/2}}{\sqrt{k+M+1}}\gamma\right)^k = 0$$

we have $P_k' \to 0$ with $k \to \infty$.

This signifies the probability that at least one point from generated by algorithm will get in the neighbourhood $O_\epsilon(z)$ tends to zero with increasing the number of iterations, which proves the Theorem. □

Theorem 4. *Assume that a positive continuous function $J(z)$ is defined in the infinite dimensional cube Π', and this function has a unique point $z^* = (x_1^*, x_2^*, \dots)$ of the global maximum. Consider the sequence σ_k given by $\sigma_k = 2R^2/\ln(k)$, where $R = \sqrt{\sum_{n=1}^{\infty} a_n^2}$. Then for any $\epsilon > 0$, the probability of choosing a point z_k from the ϵ-neighbourhood $\omega(z^*)$ of the point z^* tends to unity when $k \to \infty$. Therefore, the sequence of points $\{z_k\}$ generated by the optimisation algorithm converges in probability to the point z^* of the global maximum of the function J.*

Proof. As it was shown in the previous theorem, the set $\omega(z^*) = \{y = (y_1, \dots, y_i, \dots) : |y_i - x_i^*| \le \epsilon/(2^i), i = 1, \dots, N\}$ is a subset of the neighbourhood $O_\epsilon(z^*)$. Now we estimate the probability for the point z_{k+1}, generated at the iteration $k+1$, to get into $\omega(z^*)$.

We introduce the following notation. $J_0 = \sup_{z\in\Pi\setminus\omega(z^*)} J(z)$; I_k is the set of indexes of the points $z_i; i = 1, \dots, n$ which get into $\omega(z^*)$; $\overline{I_k}$ is the set of indexes of the points $z_{k,i}, i = 1, \dots, k$ which land outside $\omega(z^*)$. According to Theorem 3, the sequence $\{z_k\}$ is everywhere dense (with the probability of unity) in Π'. Since J is a continuous function, there will always be a point $z_p \in \omega(z^*)$ such that we have $J(z_p) > J_0$.

For $k > p$, the probability of choosing the point $z_{k,j}$ as a reference point from the set $\overline{I_k}$ can be estimated as follows.

$$\frac{\Sigma_{j\in\overline{I_k}} J^{\phi_k}(z_{k,j})}{\Sigma_{j=1}^{n} J^{\phi_k}(z_{k,j})} \le \frac{\Sigma_{j\in\overline{I_k}} J_0^{\phi_k} J^{-\phi_k}(z_p)}{\Sigma_{j=1}^{n} J^{\phi_k}(z_{k,j}) J^{-\phi_k}(z_p)} < n\Big(\frac{J_0}{J(z_p)}\Big)^{\phi_k} \xrightarrow{k\to\infty} 0$$

Since this probability tends to zero, then the probability of choosing the point z_j as a reference point z' with the index j from the set I_k tends to unity. We estimate the probability $P_{k+1}(\omega(z^*))$ of choosing the point z_{k+1} of trial at $(k+1)$th iteration of the optimisation algorithm, under condition that the reference point z' for the given iteration is chosen from the set I_k, i.e. $z' \in \omega(z^*)$.

$$P_{k+1}(\omega(z^*)) = \prod_{i=1}^{N} \frac{\int_{x^*-\epsilon/(2^i)}^{x^*+\epsilon/(2^i)} \exp\Big(-\frac{(y_i-x_i')^2}{s_{k+1,i}}\Big) dy_i}{\int_{-a_i/2}^{a_i/2} \exp\Big(-\frac{(y_i-x_i')^2}{s_{k+1,i}}\Big) dy_i}$$
$$\ge \prod_{i=1}^{N} \Big[\Phi\Big(\frac{x_i^*+\epsilon/(2^i)-x_i'}{s_{k+1,i}}\Big) - \Phi\Big(\frac{x_i^*-\epsilon/(2^i)-x_i'}{s_{k+1,i}}\Big)\Big],$$

where the function Φ is the Gauss error function:

$$\Phi(t) = \frac{2}{\sqrt{\pi}} \int_0^t \exp(-\tau^2) d\tau.$$

Since $z' \in \omega^*$, we have: $x_i^* - \epsilon/(2^i) < x_i' < x_i^* + \epsilon/(2^i)$, and $x_i^* + \epsilon/(2^i) - x_i' > 0$, $x_i^* - \epsilon/(2^i) - x_i' < 0$.
Since $1/s_{k+1,i} \to +\infty$ for $k \to +\infty$, then

$$\Phi\Big(\frac{x_i^*+\epsilon/(2^i)-x_i'}{s_{k+1,i}}\Big) \to 1, k \to +\infty; \Phi\Big(\frac{x_i^*-\epsilon/(2^i)-x_i'}{s_{k+1,i}}\Big) \to 0, k \to +\infty.$$

Therefore, from the obtained above inequality for $P_{k+1}(\omega(z^*))$, it follows $P_{k+1}(\omega(z^*)) \to 1$, for $k \to \infty$.

This signifies that the probability of choosing a point from the set $\omega(z^*)$, and consequently from the neighbourhood $O_\epsilon(z^*)$, tends to one for a gradual increase of the iteration number k. This completes the proof of the theorem, and demonstrates the convergence of the optimisation algorithm in Hilbert spaces. □

Appendix B

Here we present a pseudocode of our basic optimisation algorithm (M_0) (it is denoted as Algorithm 1). Algorithm 1 uses various functions, which are listed below as Algorithms 2, 3, 4, and 5. The detailed description of (M_0) can be found in Section 2. The parameters m, a, n, Q, M have the following meaning: m is the dimensionality of space; a symbolically denotes the cube with sides $[-a, a]$, where the search takes place (for non-symmetric cases, the extension of the code is straightforward); n is the size of the population of points; Q and M are parameters of the method, defined in Section 2. Algorithms 4 and 5 use sequences of numbers ϕ_k and σ_k, which should be predefined. The function 'Find_Index_of_Max(population_J)' is a standard function, which finds the index of the element in an array having the maximal value; it is not listed here. The function 'normal_distribution (mean,variance)' generates a normally distributed random number.

Algorithm 1 Basic Optimisation Algorithm (M_0)

```
Require: m, a, n, Q, M, number_of_iterations, Objective_Function;
  population ← Generate_Population(m, a, n);
  population_J ← Calculate_J(population, Objective_Function);
  for k := 1 → number_of_iterations do
      (selected_point, selected_index) ← Selector(k, population, population_J);
      mutated_point ← Mutatuion_Operator(k, a, Q, M, selected_point);
      current_J ← population_J[selected_index];
      new_J ← Objective_Function(mutated_point);
      if new_J ≥ current_J then
          population[selected_index] ← mutated_point;
          population_J[selected_index] ← new_J;
      end if
  end for
  max_index ← Find_Index_of_Max(population_J);
  return population[max_index];
```

Algorithm 2 Generate_Population

```
Require: m, a, n;
  for i := 1 → n do
      for j := 1 → m do
          current_point[j] ← rand(−a, a);
      end for
      population[i] ← current_point;
  end for
  return population;
```

Algorithm 3 Calculate_J

```
Require: population, Objective_Function;
  for i := 1 → size(population) do
      population_J[i] ← Objective_Function(population[i]);
  end for
  return population_J;
```

Algorithm 4 Selector

```
Require: k, population, population_J;
  for i := 1 → size(population) do
      probability[i] ← population_J[i]^{Φ_k};
  end for
  max ← probability[1];
  for i := 1 → size(population) do
      if probability[i] > max then
          max ← probability[i];
      end if
  end for
  sum ← 0;
  for i := 1 → size(population) do
      probability[i] ← probability[i]/max;
      sum ← sum + probability[i];
  end for
  for i := 1 → size(population) do
      probability[i] ← probability[i]/sum;
  end for
  probability_interval[1] ← 0;
  for i := 2 → size(population) + 1 do
      probability_interval[i] ← probability_interval[i − 1] + probability[i − 1];
  end for
  r ← rand(probability_interval[1], probability_interval[size(population) + 1]);
  index ← 1;
  for i := 2 → size(population) + 1 do
      if probability_interval[i] ≥ r then
          index ← i − 1;
          break
      end if
  end for
  return (population[index], index);
```

Algorithm 5 Mutation_Operator

Require: $k, a, Q, M, selected_point$
  $m \leftarrow size(selected_point)$;
  for $i := 1 \rightarrow m$ **do**
    if $rand(0,1) < Q$ **then**
      $mutated_gene \leftarrow normal_distribution(selected_point[i], \sigma_k/2)$;
    else
      $mutated_gene \leftarrow normal_distribution(selected_point[i], \sigma_{k+M}/2)$;
    end if
    if $mutated_gene > a$ **then**
      $mutated_gene \leftarrow a$;
    end if
    if $mutated_gene < -a$ **then**
      $mutated_gene \leftarrow -a$;
    end if
    $mutated_point[i] \leftarrow mutated_gene$;
  end for
  return $mutated_point$;

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